

Part-A

1. Define Capacity constraint and Flow conservation for a Network flow graph?
2. What are Approximation algorithms?
3. What is an optimal binary search tree?
4. Define P and NP Class problem.
5. State Cook's theorem.
6. Define the terms NP-complete and NP-hard
7. List the Class of problems beyond NP – P SPACE
8. Define Residual Network with an example.
9. Define augmenting path with an example.
10. Define tractable and intractable problems.
11. Write control abstraction for backtracking strategy.
12. State the need of explicit and implicit constraints in backtracking algorithm?

Part-B

1. Explain the solution to the 8-queens problem using backtracking
2. Explain Branch and Bound algorithm for Travelling salesperson problem.
3. Explain 3-SAT problem and prove that 3-SAT is NP complete.
4. Explain the significance of randomization in algorithms.
5. Explain topological sorting with an example.
6. Write a short note on Approximation algorithms?
7. Explain the process involved to show that a problem is NP-complete.
8. Explain Polynomial time Approximation schemes with an example.
9. Compare Branch and-Bound and Backtracking strategies for the design of solution for Knapsack Problem
10. Using Ford-Fulkerson Method, compute maximum flow for the given network.

